Storm Warning: Can Weather Data Save Virginia's Power Grid?

DS 4002 Case Study by Dino Papalabrakopoulos

Electricity demand fluctuates daily and seasonally due to behavioral, economic, and environmental factors. Among these, weather is one of the most influential variables, as temperature extremes significantly increase heating and cooling demand in residential and commercial sectors. With climate variability increasing and the energy demands of the modern digital economy growing – driven in large part by the rapid proliferation of data centers across the Mid-Atlantic region – understanding the relationship between weather and electricity demand has become increasingly critical. Dominion Energy, a major provider in this region, has cautioned regulators that demand growth is outpacing new generation capacity, underscoring the need for more accurate forecasting tools to support energy resilience and sustainability planning.

Utility providers frequently rely on statistical forecasting methods to anticipate short-term load demand, prevent blackouts, and optimize resource allocation. Conventional forecasting models have underutilized environmental variables, relying instead on historical demand observations that cannot fully capture the influence of weather on consumption. Temperature and seasonal cycles are strong predictors of electricity demand, suggesting that integrating weather observations with historical load data could meaningfully improve forecast accuracy. As demand patterns grow more complex – shaped by seasonal weather cycles, data center usage patterns, and other behavioral factors – this omission becomes increasingly consequential. Developing forecasting approaches that incorporate environmental variables alongside historical demand represents a promising avenue for improving grid reliability and contributing to broad discussions about climate-sensitive infrastructure planning.

You are a data scientist contracted by a regional grid planning office to evaluate whether incorporating weather data improves short-term electricity demand forecasts for the Dominion load area of Virginia. The wake-up call came this past winter, when a bomb cyclone swept across the Mid-Atlantic and sent electricity demand surging to levels the old forecasting models never saw coming. Operators scrambled to alleviate grid stress, and, in some cases, customers lost power. Senior engineers have since flagged that the model is systematically blind to the kind of extreme weather swings that are becoming less and less rare. With summer peak season now approaching and forecasts already foreboding, leadership needs an answer before the next crisis hits. You will build and evaluate two competing forecasting models, put them head-to-head, and deliver a clear, evidence-based verdict on whether weather data belongs in the forecast – one that grid operators can act on before the next extreme weather event puts the lights at risk.